
EXERCISE

Lecture 6: Auction Activity

AI508 — Algorithmic Game Theory

Simon Holm
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1 Revenue Maximization

Recall welfare and utility:

$$\text{welfare} = \sum_{i=1}^n v_i x_i, \quad \text{utility} = v_i x_i - p_i$$

Now note sellers revenue

$$\text{sellers revenue} = \sum_{i=1}^n p_i$$

And that for *total utility* (including seller)

$$\sum_{i=1}^n v_i x_i - p_i - \sum_{i=1}^n p_i = \sum_{i=1}^n v_i x_i$$

How can one maximize seller revenue?

1.1 Simplest case

Assume 1 item and 1 bidder.

Let the single bidders valuation $v \sim F$ (continuous on has a density f)

$$\text{supp}(F) = [0 \leq v \leq \bar{v}]$$

We will sell the item if and only if $v \geq r$ and the bidder pays r .

$$\mathbb{P}(\text{making a sale}) = \mathbb{P}(v \geq r) = 1 - F(r)$$

Where $F(x) = \int_{-\infty}^x f(x) dx$

Then the sellers revenue is $r(1 - F(r))$

Then

$$r^* = \arg \max_r r(1 - F(r))$$

$$\frac{\partial}{\partial r} r(1 - F(r)) = 1 - F(x) - rF'(x) = 0$$

$$\implies r = \frac{1 - F(r)}{F'(x)} = \frac{1 - F(r)}{f(x)}$$

1.2 Virtual welfare

Add some lower bound for the bids such that

$$\text{Welfare} = \sum_{i=1}^n \phi(v_i) x_i$$

Where

$$\phi \stackrel{\text{def}}{=} v - \frac{1 - F(v)}{f(v)} \text{virtual welfare}$$

v^* is a point where $\phi(v^*) = 0$ is the revenue price.